

Comparative Analysis and Performance Evaluation of Classification Models for Diabetes Prediction

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1. Abstract

This project focuses on the prediction of diabetes using machine learning classification techniques on the diabetes.csv dataset. The dataset contains medical diagnostic features such as glucose level, blood pressure, insulin, BMI, age, and diabetes pedigree function, with the target variable indicating whether a patient is diabetic or non-diabetic. The main goal of the project was to compare the performance of multiple classification algorithms and identify the most suitable model for diabetes prediction. The project included fitting data, data analysis, train-test splitting, model training, and performance evaluation. Several classification models such as Logistic Regression, K-Nearest Neighbors, Support Vector Machine were applied. The models were evaluated using ROC_AUC score. Among the tested models, the support vector machine produced the best performance with maximum roc_auc_score which is 0.6742. The project demonstrates how machine learning can support early diagnosis of diabetes and assist in healthcare decision-making. It demonstrates that careful data preparation and comparing different machine learning models are key to achieving better prediction results.

2. Introduction

Diabetes is one of the most common chronic diseases worldwide and early prediction is important for timely diagnosis and treatment. The increasing availability of healthcare data has made machine learning an effective approach for disease prediction and classification problems. In this project, a diabetes prediction system was developed using a structured medical dataset containing patient-related health indicators such as pregnancies, glucose, blood pressure, skin thickness, insulin, BMI, diabetes pedigree function, and age. The target variable in the dataset indicates whether a patient is diabetic or not.

The project is relevant because it applies data science and machine learning techniques to a real-world healthcare problem. The technologies involved in this work include Python, Google Colab, and machine learning libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. The project began with loading the dataset from Google Drive in Colab, followed by data understanding, preprocessing, and model building. We compared several classification algorithms to see how well each one could predict outcomes. The purpose of doing this project was to understand the complete machine learning workflow, from data collection and cleaning to model evaluation and interpretation. It also helped in learning how different algorithms behave on the same healthcare dataset and how roc_auc_scores can be used to select the best-performing classifier.

list of topics that you received training on during the first two weeks of internship:

1. Python programming
2. Working with Google Colab
3. Numpy and pandas
4. ROC AUC curve
5. Data visualization using Matplotlib

6. Basics of statistics for data science
7. Supervised machine learning
8. Classification algorithms
9. Model evaluation metrics
10. Train-test splitting
11. LLM models
12. Communication skills

3. Project Objective

The main objectives of this project are as follows:

1. To build a machine learning based system for predicting whether a person is diabetic or non-diabetic.
2. To analyze the diabetes dataset and understand the effect of medical features on the target outcome.
3. To train and compare multiple machine learning classification algorithms on the same dataset.
4. To evaluate the models using performance metrics such as accuracy, precision, recall, F1 score and roc_auc_score.
5. To identify the best-performing classification model for diabetes prediction.
6. To understand the practical application of predictive analytics in the healthcare domain.

4. Methodology

The project was carried out using a structured machine learning workflow in Google Colab. The dataset used for the project was diabetes.csv, which was stored in Google Drive and loaded into the Colab environment for analysis. The dataset consists of 768 records and 9 columns, including 8 prediction variables and 1 target variable named Outcome.

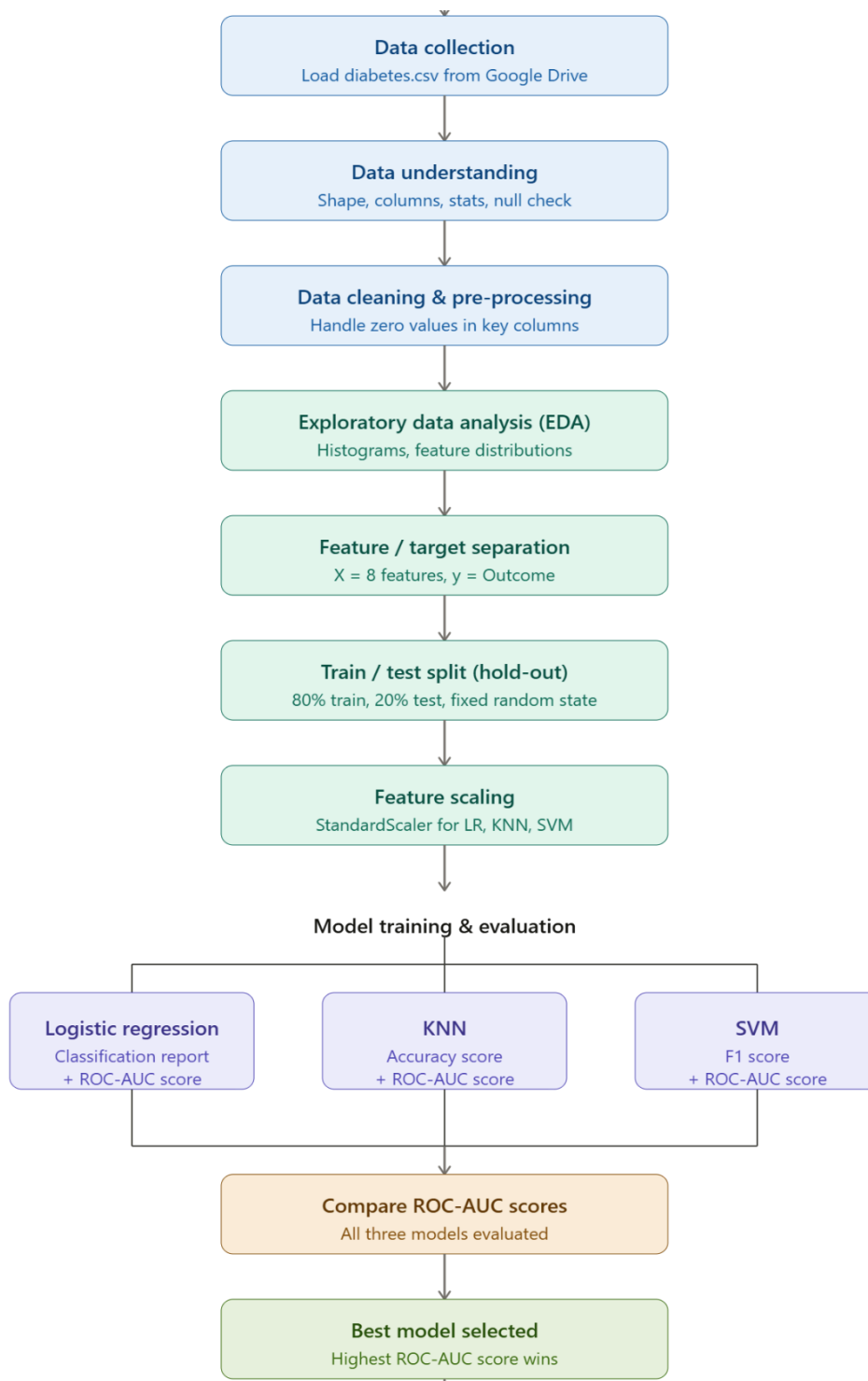
The first step was data collection, the dataset was accessed from Google Drive. After loading the dataset, the data structure, column names, null values, data sets shape, and statistics were examined to understand the dataset properly. The next step involved data cleaning and pre-processing. During this stage, the dataset was checked for missing values. In this dataset, some columns such as glucose, blood pressure, skin thickness, insulin, and BMI contain zero values, which may represent invalid or missing medical observations. These values were considered during analysis.

After cleaning, exploratory data analysis was performed to better understand the distribution of features and their relationship with the target variable through Histograms. For model development, the input features were separated from the target variable. The dataset was then divided into training and testing subsets using an 80 percent training and 20 percent testing split. A fixed random state was used to ensure reproducibility. For algorithms sensitive to

feature scale, such as Logistic Regression, KNN, and SVM, feature scaling was applied using StandardScaler.

Several classification algorithms were trained and tested such as Logistic Regression, K-Nearest Neighbors, Support Vector Machine. We chose the best model by comparing how well each one made predictions on the test data. To check this, we used a hold-out method, splitting the data into training and testing sets.

Logistic regression has done classification reports, KNN has done accuracy and SVM has done the F1 score. Finally each model has evaluated their roc_auc_score through which they are compared. We use ROC-AUC score because it measures how well a classification model can distinguish between classes, not just how many predictions it got correct. After comparing all models, the best-performing model was identified based on maximum roc_auc_score.



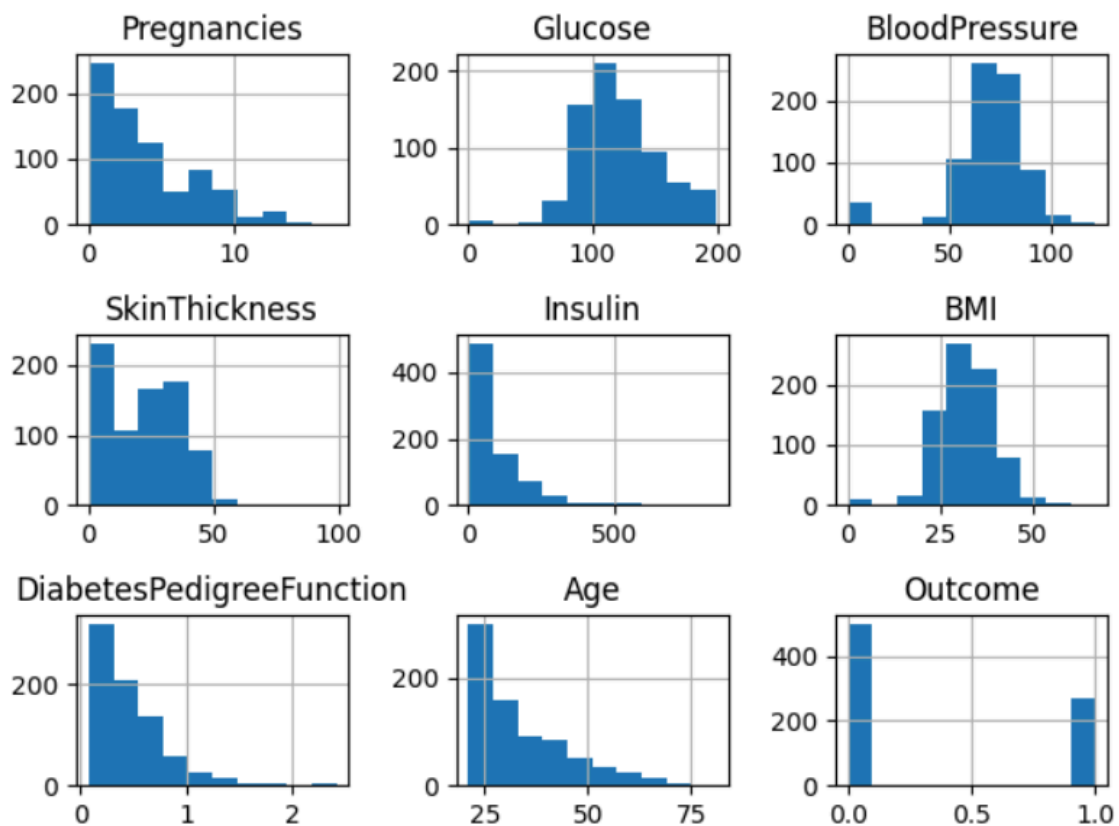
GIT HUB LINK: <https://github.com/vyshnavisur2008-tech/TIH-project>

5. Data Analysis and Results

Descriptive Analysis

The diabetes dataset contains 768 observations and 9 variables. Out of these, 8 are independent variables and 1 is the dependent target variable. The target variable Outcome indicates whether a patient is diabetic or not. The dataset distribution shows 500 non-diabetic cases and 268 diabetic cases, indicating that the majority class is non-diabetic.

shape of the datasets distribution:



The average values of important features are approximately:

	Pregnancies	Glucose	Blood pressure	Skin thickness	Insulin	BMI	Diabetes pedigree function	Age
Mean	3.85	120.89	69.11	20.54	79.80	31.99	0.47	33.24

The data also reveals the presence of zero values in some medical measurements, which indicates the need for preprocessing attention. Feature visualization and summary statistics help understand the spread, and possible outliers in the data.

Inferential / Predictive Analysis

To perform predictive analysis, multiple machine learning classification algorithms were trained and tested. The dataset was split into training and testing sets using an 80:20 ratio.

The comparative model evaluation results are shown below:

Model	SVM	Logistic Regression	KNN
ROC-AUC score	0.6742592592592592	0.6692592592592593	0.6592592592592593

From the results, SVM achieved the ROC-AUC value , while KNN has the lowest score so SVM can be considered the best-performing model for this project. This comparison helped in understanding which machine learning algorithm is more suitable for diabetes prediction using this dataset.

6. Conclusion

This project successfully demonstrated the use of machine learning techniques for diabetes prediction using the diabetes.csv dataset. The work included data loading, preprocessing, exploratory analysis, classification model development, and comparative evaluation. The dataset consisted of 768 patient records with health-related features that were used to predict diabetes outcome.

The comparative study showed that different algorithms perform differently on the same dataset. Among the evaluated models, SVM delivered the highest ROC AUC value of approximately 0.6742, making it the most suitable model in this analysis. The project highlights the practical importance of machine learning in healthcare and how predictive models can support early disease identification. For future work, the project can be extended by handling zero values more carefully as missing values, performing hyperparameter tuning, applying cross-validation, balancing the dataset, and testing advanced models such as XGBoost. Model deployment through a simple web app can also be considered to make the system more usable in real-world scenarios.

7. APPENDICES

1. Dataset: [diabetes.csv](#)
2. Github link: <https://github.com/vyshnavisur2008-tech/TIH-project>
3. Diabetes report: [report](#)

